Attitude adjustment:
Trends in the semantics of clausal-embedding

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Day 3

Outline

Bits from yesterday

Today

2. How [the truth conditions of belief sentences] are compositionally derived

We have disregarded some facts...

Recap

Propositions:

(1) $[\![\text{it's raining}]\!] = \{w: \text{ it's raining at } w\}$

(2) $[\![\text{it's raining and Brno is in Austria}]\!] =$
$\{w: \text{ it's raining at } w\} \cap \{w: \text{ Brno is in Austria at } w\}$

Individuals' doxastic alternatives:

(3) $\text{DOX(a)} = \{w : w \text{ is compatible with a's beliefs (at } w_0)\}$

(4) $\text{DOX(a)} = \bigcap\{p: \text{ a believes } p \text{ (at } w_0)\}$

The truth conditions of "a believes p" per Hintikka

(5) $\text{DOX(a)} \subseteq \{w: \text{ it's raining at } w\}$

Recap

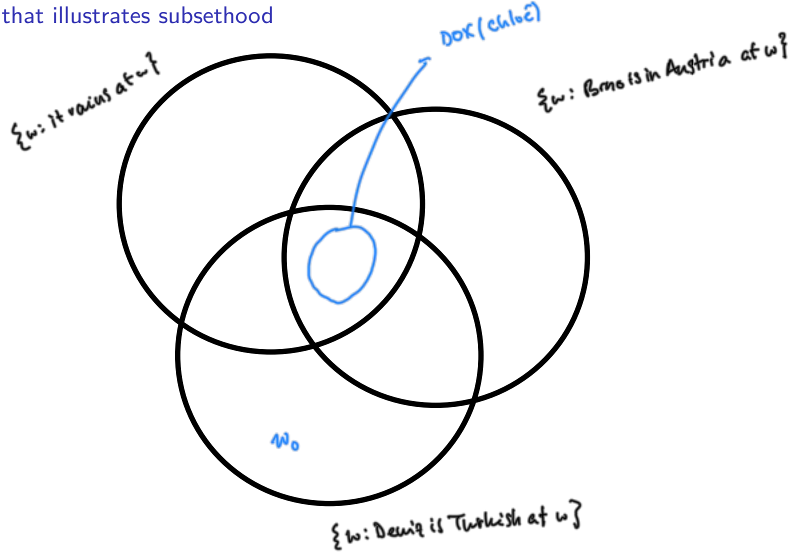
picture that illustrates subsethood

Take three propositions: rain, brno, turkish

- What makes “Chloé believes that it’s raining” true?
- Where is the actual world?

Recap

picture that illustrates subsethood



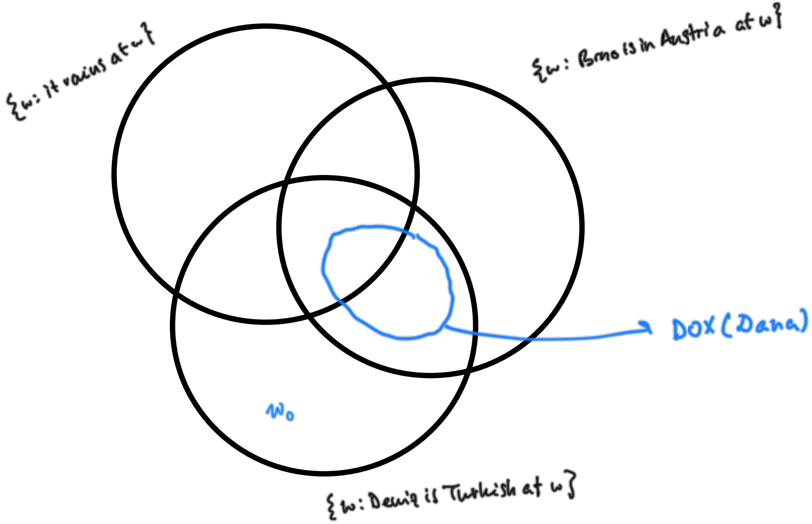
Recap

picture that illustrates non-subsethood

Assume that Dana has the same beliefs as Chloé except that she's not opinionated as to whether it's raining. What does her belief set look like?

Recap

picture that illustrates non-subsethood



Recap

Take away:

all of DOX are p worlds: x believes p

some are p some are not-p worlds: x is unopinionated wrt p

none of DOX are p worlds: x believes not-p

Equality or subsethood?

Take two of Chloé's beliefs:

(6) a. She believes that it's raining.
    b. She believes that Brno is in Austria.

If our semantics for belief *equated* DOX(Chloé) with these two propositions, we would get:

(7) a. $\text{DOX(Chloé)} = \{w : \text{ it's raining at } w\}$
    b. $\text{DOX(Chloé)} = \{w : \text{ Brno is in Austria at } w\}$

This entails (8), which problematic.

(8) $\{w : \text{ it's raining at } w\} = \{w : \text{ Brno is in Austria at } w\}$

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This entails (8), which problematic—but not quite enough.

(8) $\{w: \text{ it's raining at } w\} = \{w: \text{ Brno is in Austria at } w\}$

We can imagine that it rains iff Brno is in Austria...?

Note to self

I was tempted to further the problem by considering:

(9) a. contingent propositions with asymmetric entailment:
buy a red car $\rightarrow$ buy a car

b. mathematical truths with asymmetric entailment:
12 is divisible by 4 $\rightarrow$ 12 is even

c. Chloé acquiring a new belief:
This is an impossible situation as her DOX was both not equal (past) and equal (now) to the new proposition...

But this is not satisfactory either:

(9a) suffers from the world knowledge problem.

how to do entailment for mathematical truths if all of them are true all across Ω?

(9c) requires us to introduce new machinery: beliefs at *times*.

Equality or subsethood

better

Take Dana, who believes that it's raining but that Brno is not in Austria.

We can construct a context in which Chloé has the beliefs that she has and Dana hers.

An equality semantics for Dana would inevitably lead to:

(10) a. $\text{DOX(Dana)} = \{w: \text{ it's raining at } w\}$
b. $\text{DOX(Dana)} \neq \{w: \text{ Brno is in Austria at } w\}$
c. $\{w: \text{it's raining at } w\} \neq \{w: \text{Brno is in Austria at } w\}$

Doing the same for Chloé would results in the equality from (8), which contradicts (10c).

Equality?

Regardless, there are people who believe in an equality semantics in refined versions of the semantics that we're sketching out.

They wouldn't deny any of the above, but given the assumptions they make, equality might be an option (Bondarenko 2022 and references therein).

Someone who believes no propositions...

Take Ω as our universe. Why does $\bigcap\{\} = \Omega$?

I tried at a proof myself and then checked on stackexchange. I can't guarantee what a mathematician would say, but I pumped Idea 1 (below) directly from there and Idea 2, what I thought, seemed approximately right. This will give you an intuition but is not rigorous. I urge you to do better than me :)

Idea 1:

Take a set of sets S. The more elements S has, the littler $\bigcap S$ will be.

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Take a set of sets S. The more elements S has, the littler $\bigcap S$ will be. The fewer elements S has, the greater $\bigcap S$ will be. And the greatest possible set is Ω.Therefore $\bigcap S = \Omega$.

Recall that $p \to q$ is true whenever p is false.

p	q	p → q
T	T	T
T	F	F
F	T	T
F	F	T

Idea 2:
What does computing $\bigcap S$ tell us to do?

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In intuition, it goes something like:

$$\text{if } X, Y \in S, \text{ then, if } x \in \cancel{X} \text{ and } x \in Y, \text{ then } x \in \bigcap S.$$

When S is not empty,

(11) $\quad \mathsf{S} = \{w1, w2\}, \{w2\}$

you get an intuitive result:

(12)
a. if X and Y are in S ('they are'), then
b. if x is in X and x is in Y ('one is'),
c. then x is in bigcup S.

So

(13) $\quad \bigcup \mathsf{S} = \{1\}$

The same reasoning will work for any non empty S.

So will it for any *empty* S.
Take S $= \{\}$

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a. if X and Y are in S ("they aren't!"), then
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The solution lies in the fact that the antecedent of the conditional is false, that we can derive whatever we want from it, but we specifically want to derive $\bigcup S = \Omega$ from it, and I (personally) don't know how to constrain things.

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2. How [the truth conditions of belief sentences] are compositionally derived

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The truth conditions of "a believes p" per Hintikka (repeated from above)

(5) $\text{DOX(a)} \subseteq \{w: \text{ it's raining at } w\}$

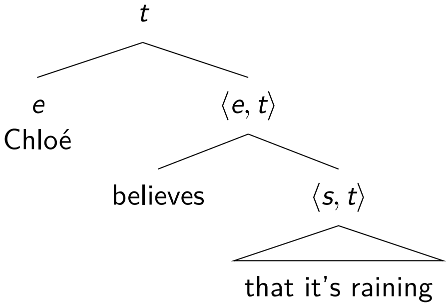
These global truth conditions for belief reports are traditionally presented together with a syntactic hypothesis.

In the trees below, what's important is the assumption that:

- the clause is the direct object of 'believe' — syntax
- the denotation of the clause is the argument of the function denoted by 'believe' — semantics
- syntactic complements are interpreted via function application

Getting to these truth conditions compositionally

types



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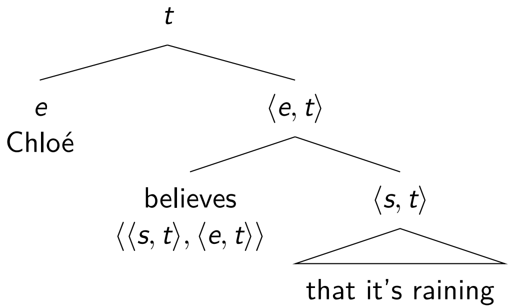

Getting to these truth conditions compositionally

the function

1 iff DOX(*Chloé*) $\subseteq \{w: \text{ it's raining at } w\}$

- *Chloé*
  Chloé
- ?
  - believes
  - $\{w: \text{ it's raining at } w\}$
    - that it's raining

Getting to these truth conditions compositionally

the function

1 iff DOX(*Chloé*) $\subseteq \{w:$ it's raining at $w\}$

Chloé
Chloé

?

believes

$\{w:$ it's raining at $w\}$

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$[\![\text{believes}]\!] = \lambda p_{\langle s,t \rangle}.\lambda x_e.$ 1 iff $DOX(x) \subseteq p$

Getting to these truth conditions compositionally

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Chloé Chloé

λx_e. 1 iff $DOX(x) \subseteq \{w : \text{it's raining at } w\}$

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that it's raining

$[\![\text{believes}]\!] = \lambda p_{\langle s,t \rangle} . \lambda x_e .$ 1 iff $DOX(x) \subseteq p$

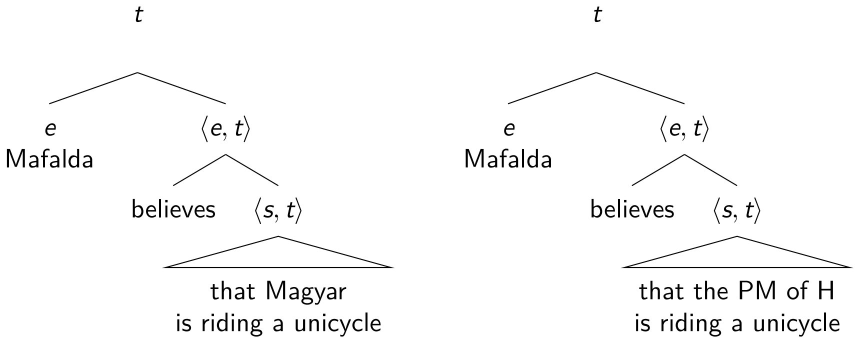
One of our problems was that if ‘believe’ were a function of type $\langle t, \langle e, t\rangle\rangle$, it would map two sentences with the same truth value on to the same output.

While these two sentences are true at our world (now), they do not have to have the same intension—they correspond to two distinct propositions.

(15) a. Péter Magyar is riding a unicycle
b. The Prime Minister of Hungary is riding a unicycle.

This is because there are worlds at which the PM of Hungary is Magyar, but others at which it is David Müller, Ari Joshi, my sister, etc.

The two propositions in the trees below are *different* sets of worlds.
'Believes' need not map onto the same object.



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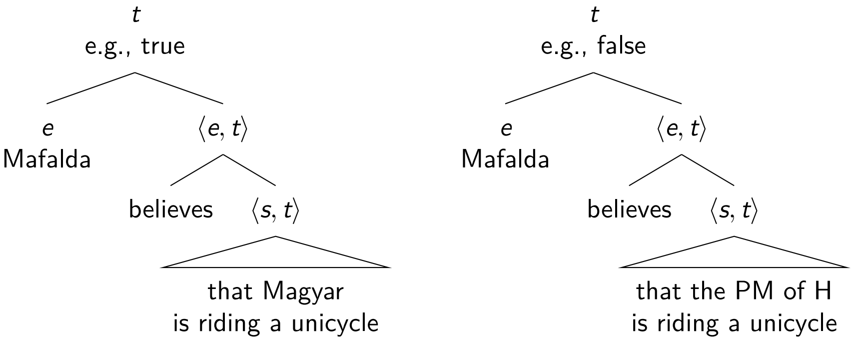

Going back to the jackalope example, we can imagine that there are worlds at which there are jackalopes.

And sentence (16) is false, but it could be that all of *Mafalda's belief worlds* have jackalopes in them.

(16) ~~Jackalope sentence~~
Jackalopes are fierce.

(16') Mafalda believes that Jackalopes are fierce.

If you're interested in further details, some places to start:

Heim & Kratzer (1998), chapter 2.

von Fintel & Heim (2011), chapters 1 & 2.

there and back again

Hintikka gave us the semantics and we fit the simplest syntax possible on to it.

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Clauses don't pattern like other direct objects

(17) a. Du hast das Haus gesehen.
you have the house seen
You have seen the house.

[compare with (18b)]

b. *Du hast gesehen das Haus.

[compare with (18a)].

(18) a. ~~Du glaubst, dass es regnet~~.
~~you believe that it rains~~
~~You believe that it's raining~~.

Du hast geglaubt, dass es regnet.
You have believed that it rains

b. ~~*Du dass es regnet glaubst~~.

*Du hast dass es regnet geglaubt.

In German, for example, clauses like to 'extrapose', whereas regular NPs stay in the middle field.

Clause-embedding verbs can combine with nouns as well as with clauses

(19) a. Chloé believes that it's raining.
b. Chloé believes the rumor.
c. Chloé believes Deniz.

Given how we defined 'believe' above, it requires its complement to be a proposition: We don't know how to combine it with 'the rumor' or 'Deniz'.

Some nouns can also combine with clauses

(20) a. the rumor that it's raining
b. *the book that it's raining

There are asymmetries in interpretation depending on these combination options

(21) a. Chloé believes that it's raining.
b. Chloé believes the rumor that it's raining.
believe the rumor p $\rightarrow$ believe p

(22) a. Chloé knows that it's raining.
b. Chloé knows the rumor that it's raining.
know the rumor p $\not\rightarrow$ know p

How do we account for this influx of data given the nice system that we have received?

Recall, we want our formal system to match

(a) what language does (e.g., nouns? nouns),
(b) the truth conditions that we derive to match our intuitions,
(c) for all of this to match the syntax of natural language.

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we have some work to do...